SOLUTION PITCH | GREAT INDIAN AI INTERNSHIP CHALLENGE

# MythosConnect

Voice-first access to knowledge in Indian languages, built on Gnani Prisma and Timbre

Aahant Kumar | Delhi Technological University (DTU)

## The problem

Most digital knowledge, from cultural content to civic information, is written in English or a few major languages and expects typed input. People who are low-literacy, elderly, or simply most comfortable speaking a regional language are left out. Cross-language access is rarer still: asking in Hindi and getting the answer in Tamil.

## The solution

MythosConnect is a voice agent. You speak a question in one Indian language and hear the answer in another, with no typing and no reading. **Speak** → **Gnani Prisma** transcribes → an LLM writes a short answer in the chosen language → **Gnani Timbre** speaks it back.

- **Three modes on one pipeline:** Mythology Guide, Student Tutor, and Citizen Helper (plain-language help with services and forms that admits uncertainty and points to official sources).
- **Independent input and output languages** across 9 Indian languages and English, plus short conversation memory for follow-up questions.
- **Observable:** per-stage latency (Prisma, LLM, Timbre) is shown for every request, and each failure names the stage that failed.

## How it uses Gnani models

- **Prisma (speech to text)** is the entry point for every request, handling recorded or uploaded audio in the user's language.
- **Timbre (text to speech)** produces the spoken reply in the target language; the app validates the audio and retries once on empty output.
- A built-in benchmark (`eval.py`) uses **both models together**: Timbre speaks test sentences and Prisma transcribes them, and the app reports word error rate and latency.

## Evidence it works

| Language | Sentences | Mean WER (%) | TTS+STT time (s) |
|---|---|---|---|
| Hindi | 3 | 8.3 | 3.6 |
| English | 3 | 5.6 | 3.4 |
| Tamil | 3 | 0.0 | 3.1 |
| Telugu | 3 | 33.3 (0.0 on successful calls) | 4.1 |

11 of 12 round trips succeeded (mean WER 3.8% on those 11). The one failure was a Telugu `EMPTY_AUDIO` error from Prisma, which led to a validate-and-retry change in the TTS wrapper. Limits: 3 sentences per language, synthetic speech, one voice; real human recordings are not yet tested.

## Impact

A student can hear an explanation in their own language. A first-time user can ask how a form works by speaking. A family can hear a story in the language the grandparents speak. Because a mode is only a prompt, new domains such as health information, agriculture or exam preparation need no new infrastructure.

## What comes next

- Replace the LLM step with **Gnani Evon** so the whole pipeline runs on Gnani models.
- Stream Timbre output to start playback earlier; add retrieval over curated sources so answers are grounded.
- Benchmark on recorded human speech and report character error rate as well as WER.